

Thomas Breydo

📞 646-262-0245 ✉ tbreydo@gmail.com [🌐 linkedin.com/in/thomas-nyc](https://www.linkedin.com/in/thomas-nyc) [🐙 github.com/thomasbreydo](https://github.com/thomasbreydo)

Education

Stanford University, CA

Sep 2023 – Jun 2027

B.S. in Computer Science – GPA: 4.21 / 4.0

- Parallel Computing, Compilers, Operating Systems, PL, Networking (CS 149, 143, 140E, 242, 144)
- Advanced Cryptography, Machine Learning, Complexity, Randomized Algorithms (CS 355, 229, 254, 265)
- Groups and Rings, Linear Algebra, Finite Fields, Probability (Math 120, 61DM, 62DM, 63DM)

Experience

Jane Street

Jun 2026 – Sep 2026

Software Engineer Intern

New York, NY

- Starting on June 15.

Xantium Group at Tudor Investment Corp.

Jun 2025 – Sep 2025

Quantitative Developer Intern

New York, NY

- Built a Python library that minimizes the annual financing cost of portfolio positions using linear programming and heuristic optimization.

Stanford AI Laboratory: Hazy Research

Apr 2025 – Jun 2025

Undergraduate Researcher

- Contributed to new theoretical work on LLM-to-LLM communication by analyzing model responses using KL divergence, mutual information, and entropy.

Stanford CS Theory Group

Jun 2024 – Sep 2024

Algorithms Research Intern

- Created custom C++ machine learning pipelines to find Nash equilibria in small bracket tournaments.
- Advised by professors Aviad Rubinstein and Moses Charikar.

Gordian Software (YC W19)

Jun 2021 – Sep 2022

Software Engineer Intern (Summers 2021 & 2022)

- Built Python data pipelines to parse airline seat maps (XML/JSON). Added support for 3 airlines.

Projects

Compiler for Custom Language in Rust [🔗](#) | Rust, LLVM, Python, Tailwind, FastAPI

- Developed a compiler (lexer, parser, code generator) for a custom language using Rust and LLVM.
- Created a sandbox using React and a FastAPI backend on AWS EC2 via Docker.

High-Performance Kernels (CS 149) | AWS NKI, CUDA, C

- Wrote a fused Conv+Bias+MaxPool kernel for AWS Trainium2 using NKI, achieving 1.4–1.8× speedup over target.
- Implemented a CUDA renderer for 2M transparent overlapping circles, running 12× faster than the reference.

Awards

Human Capital Fellowship: \$50k funding for a future startup, “no strings attached,” 2026.

USA Computing Olympiad: Gold division, 2022.

American Invitational Math Exam: Three-time national AIME qualifier, 2021-2023.

Carnegie Mellon PicoCTF Cybersecurity: Placed #18 nationally (as a team of 2), 2022.

The Dalton School: Billy Farrell Award for kindness, involvement, and character (1 recipient/year), 2023.

Technical Skills

Languages: Python, C/C++, Rust, Java, SQL, HTML/CSS/JS

Technologies: CUDA, AWS NKI, Numpy, PyTorch, Docker, Git, GitHub, Unix, Vim, LaTeX

Knowledge: Compilers, parallel computing, ML/AI, language models, operating systems, cryptography

Interests & Languages

Interests: Musical theater, guitar, teaching, linguistics

Languages: Fluent in Russian and Spanish